3.15

ADULTS

Surgical Related Infections

CHAPTER 3.15	SURGICAL RELATED INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Hepato-Pancreato-Biliary			
Acute Pancreatitis			
Mild to moderate pancreatitis including alcoholic pancreatitis	Antibiotics not required		
Severe non-necrotizing pancreatitis	Antibiotics not required		• Perform an image guided percutaneous aspiration before antibiotics.

CHAPTER 3.15	SURGICAL RELATED INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Severe non-infected necrotizing pancreatitis	Antibiotics not required		• Prophylactic antibiotics to prevent pancreatic infection are no longer recommended because of lack of benefit and possibly harm. • Surgical debridement is the treatment of choice where clinically indicated.
Severe infected necrotizing pancreatitis: infected pseudo cyst, pancreatic abscess Enterobacteriaceae Enterococci *Staphylococcus aureus* Streptococcus *Staphylococcus epidermidis* Anaerobes Candida spp. (rarely)	Piperacillin/Tazobactam 4.5 g IV q6H	Cefepime 2 g IV q8H + Metronidazole 500 mg IV q8H	• To obtain cultures (preferable tissue culture/ infected drain fluid), streamline antibiotics once culture and sensitivity result is available. • May consider meropenem if ESBL is suspected. • Duration depending on the organ involved and type of extra-pancreatic infection or collection which may require operative debridement of percutaneous drainage. If culture is negative and no source of infection identified, will need to stop the antibiotics.
Biliary Tract Infection			

CHAPTER 3.15	SURGICAL RELATED INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Cholecystitis, cholangitis or biliary sepsis secondary to common duct obstruction, gallbladder empyema Enterobacteriaceae (>50%) Bacteroides spp. (4-20%)	Co-amoxiclav 1.2 g IV q6H for 4-7 days	Ampicillin/Sulbactam 1.5-3 g IV q6-8H **OR** Cefazolin 2 g IV q8H + Metronidazole 500 mg IV q8H	
If hospital-acquired	Piperacillin/Tazobactam 4.5 g IV q6H	Cefepime 2 g IV q8H + Metronidazole 500 mg IV q8H	
Gastrointestinal			
Diverticulitis			
Antibiotics are generally not indicated, but considered for patients with the following: Fever, elevated WBC, patients who have failed to respond to conservative management, peritonitis, age > 70, immunosuppression and significant co-morbidity.			

CHAPTER 3.15	SURGICAL RELATED INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Community-onset, no prior antibiotic use for this episode	Co-amoxiclav 1.2 g IV q8H	Cefuroxime 1.5 g IV q8H + Metronidazole 500 mg IV q8H	• Total duration: 7-10 days. • Deescalate antibiotic once culture and sensitivity result is available. • IV antibiotics should be de-escalated to oral once clinical improvement occurs (refer to IV to oral conversion section in 'Principle of Antibiotic Prescribing' chapter).
Hospital-onset	Ceftazidime 2 g IV q8H **OR** Cefepime 2 g IV q8H + Metronidazole 500 mg IV q8H		
Hospital-onset, severe	Piperacillin/Tazobactam 4.5 g IV q6H		
Gastrointestinal Tract Perforation			

CHAPTER 3.15	SURGICAL RELATED INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Community-onset	Co-amoxiclav 1.2 g IV q6-8H	Cefuroxime 1.5 g IV q8H + Metronidazole 500 mg IV q8H	• Duration 4-7 days (provided there is adequate source control, no delay in surgical intervention and the patient has rapid clinical recovery).
Hospital-onset	Ceftazidime 2 g IV q8H **OR** Cefepime 2 g IV q8H + Metronidazole 500 mg IV q8H		
Hospital-onset, severe	Piperacillin/Tazobactam 4.5 g IV q6H		
Appendicitis			
Immediate appendectomy, as opposed to interval appendectomy, should be performed within 12 hours of decision to operate, except in the case of an unstable/septic patient or the presence of free perforation or generalised perforation, where surgery should be performed emergently.			

CHAPTER 3.15	SURGICAL RELATED INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Simple appendicitis, non-perforated appendicitis, perforated appendix, perforated viscus or intestine, appendicular mass Enterobacteriaceae Enterococci Bacteroides	Co-amoxiclav 1.2 g IV q6-8H	Cefuroxime 1.5 g IV q8H + Metronidazole 500 mg IV q8H	• Treatment should be discontinued within 24 hours if no evidence of perforation, abscess, or local peritonitis.
Rectal Abscess			
Need to rule out sexually transmitted infections.			
Rectal abscess	Co-amoxiclav 1.2 g IV q8H **OR** Ampicillin/Sulbactam 1.5-3 g IV q6-8H	Cefazolin 2 g IV q8H **OR** Cefuroxime 1.5 g IV q8H + Metronidazole 500 mg IV q8H	• Drainage is required. • Duration 4-7 days (if adequate source control, no delay in surgical intervention and patient has rapid clinical recovery). • Routine antibiotics are not recommended in otherwise healthy patients.
Trauma			
Abdominal Trauma, Stab Wound, Suspected Bowel or Solid Organ Injury			

CHAPTER 3.15	SURGICAL RELATED INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Gram-negative aerobes and anaerobes	Co-amoxiclav 1.2 g IV q6-8H	Ampicillin/Sulbactam 1.5-3 g IV q6-8H	• Duration 4-7 days (if adequate source control, no delay in surgical intervention and patient has rapid clinical recovery).
Vascular			
Mycotic Aneurysm and Vascular Prosthesis Infection			
Mycotic Aneurysm *S. aureus* Salmonella spp.	Ceftriaxone 2 g IV q24H ± Vancomycin* 15 mg/kg IV q8-12H		• Duration: at least 6 weeks (IV then oral based on clinical response and cultures). • Consider Vancomycin if suspecting MRSA/CoNS vascular prosthesis infection. • If *Burkholderia pseudomallei* is suspected, refer to **Chapter 3.14: Tropical Diseases**.
Vascular Prosthesis Infection	Piperacillin/Tazobactam 4.5 g IV q6H		
Ischaemic Limb Ulcers with Infection			
Ischaemic Limb Ulcers with Infection	Ampicillin/Sulbactam 1.5-3 g IV q6-8H	Co-amoxiclav 1.2 g IV q8H	• Duration: 7 days. • May need longer duration if bone infection and debridement for adequate source control.
Soft Tissue			

CHAPTER 3.15	SURGICAL RELATED INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Breast Abscess, Mastitis			
S. aureus **Low to moderate risk**	**Outpatient Therapy:** Cloxacillin 1 g PO q6H **OR** Cefalexin 500 mg PO q6H	**Inpatient Therapy:** Cloxacillin 2 g IV q6H **OR** Cefazolin 1-2 g IV q8H	• Drainage may be required for abscess. • For lactating mastitis: consider sending breastmilk for C&S if not responding after 48 hours of initial antibiotic therapy or recurring mastitis. • Duration: 10-14 days but may be shorter if clinically responding.
High risk (Patient with uncontrolled DM, immunosuppressive therapy, neutropenia)	**Outpatient Therapy:** Co-amoxiclav 625 mg PO q8H **OR** Ampicillin/Sulbactam 750 mg PO q12H	**Inpatient Therapy:** Co-amoxiclav 1.2 g IV q8H **OR** Ampicillin/Sulbactam 1.5-3 g IV q6-8H	

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